

TRAINING & EFFICIENCY

RoBERTa (2019)

- ADD** 10x More Data
- REM** NSP Task
- OPT** Dynamic Masking

Proven under-optimization of original BERT training.

ALBERT (2019)

- SHRINK** Factorized Embeddings
- OPT** Cross-Layer Parameter Sharing

Significant parameter reduction vs. BERT-Large.

ELECTRA (2020)

- NEW** Replaced Token Detection
- OPT** 100% Token Training Signal

Achieved competitive results at 1/4 compute cost.

CONTEXT & MODELLING

XLNet (2019)

- NEW** Permutation Language Modeling
- OPT** Pre-training/Fine-tuning Discrepancy solved

Avoids [MASK] discrepancy, bidirectional context.

Big Bird (2020)

- NEW** Sparse Attention (Global/Random/Local)
- EXTEND** 4,096 Context Window

Solved the fixed context problem using $O(N)$ complexity.

ModernBERT (2024)

- EXTEND** 8,192 Context Window
- ADD** RoPE Embeddings / Efficient Padding

Modernized encoder for long-doc and coding.

DEPLOYMENT & SCALE

DistilBERT (2019)

- COMP** Knowledge Distillation (Layer Removal)
- OPT** 60% Faster Inference

Retained ~97% of BERT-Base performance.

TinyBERT (2019)

- NEW** Two-Stage Distillation
- OPT** Attention & Hidden Layer distillation

Smaller and faster than DistilBERT, competitive accuracy.

Sentence-BERT (2019)

- NEW** Siamese Architecture
- EFF** Fixed-size meaningful embeddings

Enabled large-scale semantic search and clustering.